

Keyboard Controls

Learning Objectives

- Implement keyboard input controls for sprite movement
- Create responsive controls using arrow keys and other inputs
- Design smooth movement systems for game characters

Engage (5-7 minutes)

When driving a bullock cart, the farmer controls direction with the reins — left, right, forward. Keyboard controls in Scratch work the same way: arrow keys steer sprites like reins steer cattle, giving players control over character movement in games and animations.

Explore (10-12 minutes)

Students experiment with “when key pressed” blocks for all four arrow keys. They adjust movement amounts and directions, seeing how responsive controls feel. Like a farmer learning to guide oxen smoothly, they fine-tune controls for natural, responsive movement.

Explain (10-12 minutes)

Keyboard controls use event blocks (“when key pressed”) to trigger movement. Each arrow key maps to a direction: up (y+), down (y-), left (x-), right (x+). The change amount determines speed — small values for precise control, large values for quick movement. Like adjusting reins for different terrains, control sensitivity affects gameplay feel.

Elaborate (8-10 minutes)

Students create a character that moves smoothly with arrow keys across a farm-themed backdrop. They add boundaries to prevent leaving the stage, like fences keeping cattle in the field, ensuring controlled, purposeful movement.

Evaluate (5-8 minutes)

Can students implement all four directional controls? Do their controls feel responsive and smooth? Can they add boundaries to constrain movement?